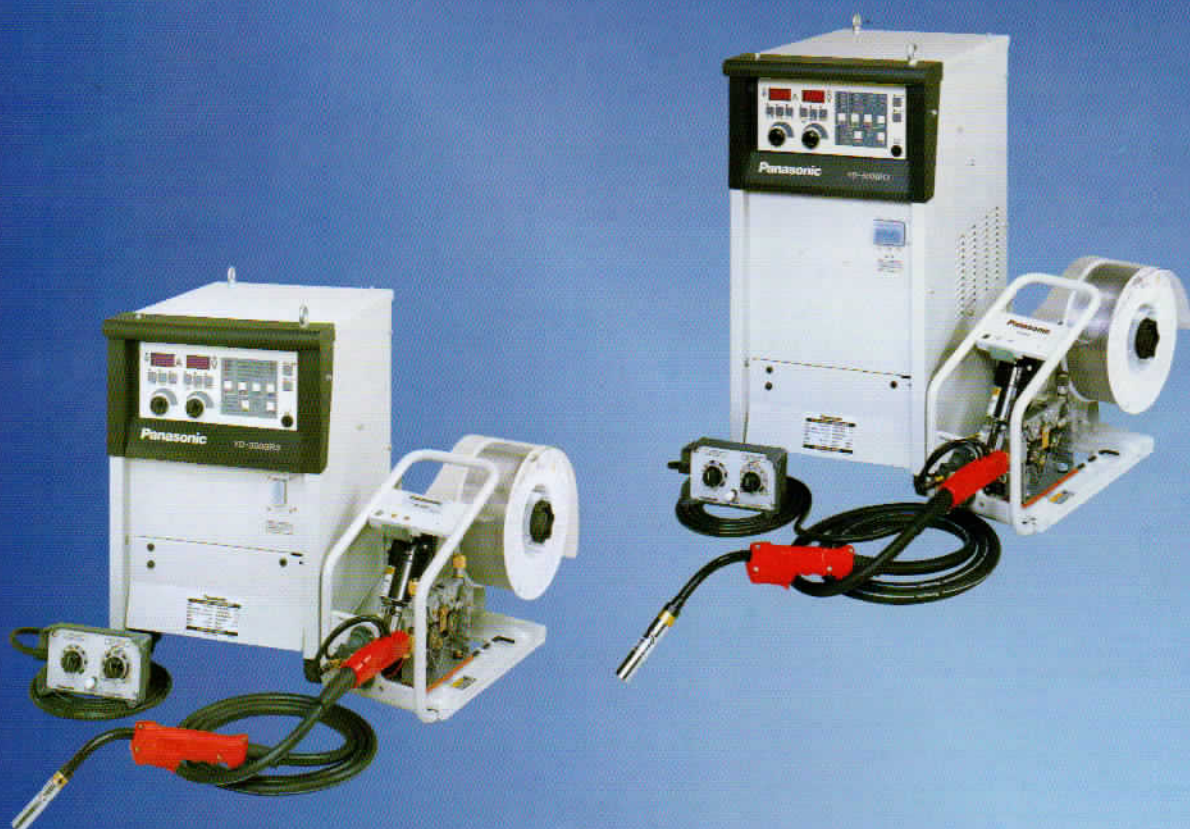


Panasonic
ideas for life

Full Digital CO₂/MAG Welding Machine

GR3 Series

Flexible welding performance and easy operation
Bring Full Digital closer to you!



GR3

Full Digital Controlled Welding Machine

FULL DIGITAL

Fusion of high grade welding of Full Digital control and simple operability of conventional analog welding machines



Comfort

- GR3 realized smooth and comfortable arc by fine control of Full Digital technology. It provides fine and stable arc by wide current range. GR3's high quality adjustment function enables fine adjustment of penetration and realizes ideal welding waveforms.

Stability on low current area (50A-120A)
(characteristic feature of Full Digital control)



Fillet weld (mild steel, 1.0mmt)
[Welding current : 80A, Welding voltage : 18V]
[Wire dia. : 1.2mm, Shielded gas : CO₂]

Improved weldability of stainless steel

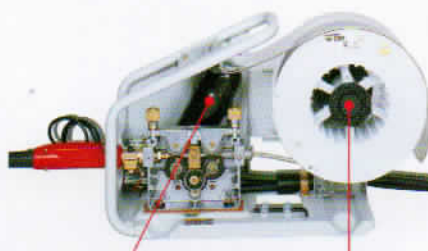


Fillet weld (stainless steel, 9.0mmt)
[Welding current : 250A, Welding voltage : 24.2V]
[Wire dia. : 1.2mm, Shielded gas : Ar + 2%O₂]

- GR3 enables quick and instantaneous arc start by starting technology in a short period of time and FTT control, which reduces the ball on the end of the welding wire at the end of the weld.

Repeatability

- Wire feed motor with encoder realize accurate wire feeding and maintains stable wire feed performance.
- GR3's set value of wire feed speed is maintained even if the primary input voltage fluctuates or the ambient temperature is changed. This allows high quality welding and makes the bead appearance straight.
- GR3 re-creates the welding conditions anywhere at anytime.



Encoder motor Wire spool shaft with brake

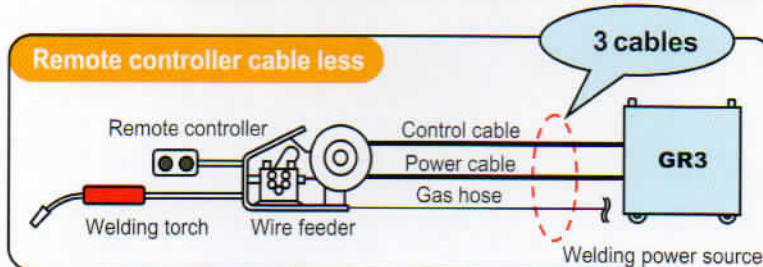
Operability

- GR3's touch panel enables easy operation. The layout of front panel is similar to conventional machines. Therefore the operator need not hustle to get used to.
- GR3 adopted same type of remote controller as conventional analog welding machines equipped.



Mobility

- GR3 has no remote controller cable between wire feeder and welding power source. This contributes to improve the working efficiency by decreasing a load on transportation. The cable is connected with wire feeder.



Friendly to the earth

- GR3 has Energy-saving function as a standard equipment. The circuit starts about 7 minutes after when the torch switch is turned off and reduces the standby energy.

- To preserve the earth environment, Panasonic Welding Systems Co., Ltd. has adopted lead free solder for the production of printed circuit boards of all the products. Also we are striving to supply



Other useful functions

1. Welding condition memory function

- GR3 can save and load the welding conditions precisely. This function is helpful to manage the welding conditions and will reduce the set up time. 9 schedules can be stored.

2. Leg length control function

- This function compensates unstable tip-to-work distance. 350GR3 keeps bead width constant and 500GR3 keeps depth of penetration constant.

Application examples



3. Abundant welding condition table

- GR3 series have abundant welding condition tables. 11 kinds of tables are equipped on 350GR3. And 10 kinds of condition tables are equipped on 500GR3.

<Table for 350GR3>

Wire diameter	Mild steel		Stainless steel	
	Solid	FCW	Solid	FCW
0.8 mm	○		○	
0.9 mm	○		○	○
1.0 mm	○		○	
1.2 mm	○	○	○	○

<Table for 500GR3>

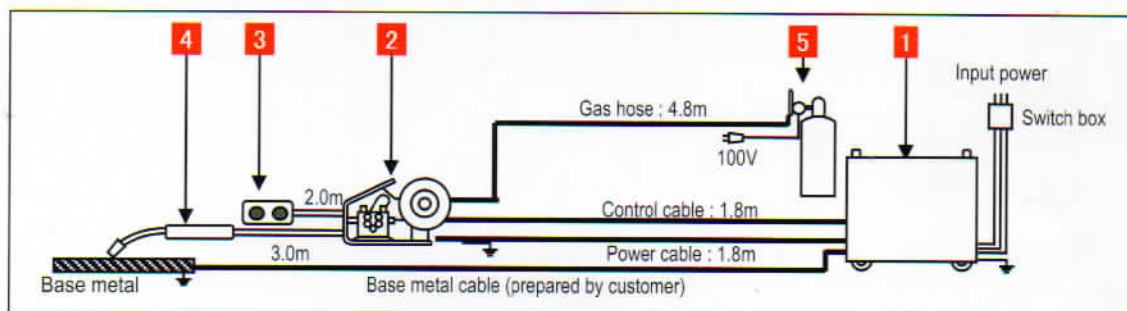
Wire diameter	Mild steel		Stainless steel	
	Solid	FCW	Solid	FCW
1.2 mm	○	○	○	○
1.4 mm	○	○		
1.6 mm	○	○	○	○

4. Robust chassis and compact design

- GR3 improved durability, dust resistance and maintenance simplicity by separating control part and main circuit in the body.

Composition of equipment

For manual welding



350GR3

500GR3

1 Welding power source

YD-350GR3Y**

YD-500GR3Y**

2 Wire feeder

YW-35DG1

YW-50DG1

3 Remote controller

YD-35GRR1

YD-50GRR1

4 Welding torch

Model (Rated current, Length, applicable dia., Duty cycle (CO₂/MAG))

•YT-18CS4	(180A, 3m, 0.8mm, 40/20%)
•YT-20CS4	(200A, 3m, 0.9/(1.0)/(1.2)mm, 50/25%)
•YT-35CS4	(350A, 3m, (0.9)/(1.0)/1.2 mm, 45/35%)
•YT-35CE4	(350A, 3m, (0.9)/(1.0)/1.2 mm, 20/20%)
•YT-35CSM4	(350A, 4.5m, (0.9)/(1.0)/1.2 mm, 45/35%)
•YT-35CEM4	(350A, 4.5m, (0.9)/(1.0)/1.2 mm, 20/20%)
•YT-35CSL4	(350A, 6m, 1.2mm, 45/35%)

Model (Rated current, Length, applicable dia., Duty cycle (CO₂/MAG))

•YT-50CS4	(500A, 3m, (1.2)/1.4/(1.6)mm, 40/35%)
•YT-50CE4	(500A, 3m, (1.2)/1.4/(1.6)mm, 25/20%)
•YT-50CSM4	(500A, 4.5m, (1.2)/1.4/(1.6)mm, 40/35%)
•YT-50CEM4	(500A, 4.5m, (1.2)/1.4/(1.6)mm, 25/20%)
•YT-50CSL4	(500A, 6m, (1.2)/1.4/(1.6)mm, 40/35%)
•YT-50CEL4	(500A, 6m, (1.2)/1.4/(1.6)mm, 25/20%)

*Optional parts are required in case of using the wire marked "()".

5 Gas regulator

- YX-25AD1 (for CO₂ / MAG / MIG)
- YX-25CC1 (for CO₂)

6 Option (Connection cable)

- YV-610GR3A / YV-615GR3A / YV-620GR3A
(Length:10/15/20m, Welding current:350A or less, Duty:60%)
- YV-310GR3A / YV-315GR3A / YV-320GR3A
(Length:10/15/20m, Welding current:250A or less, Duty:50%)

- YV-810GR3A / YV-815GR3A / YV-820GR3A
(Length:10/15/20m, Welding current:500A or less, Duty:50%)

(Note) Model numbers and specifications are subject to change without notice.

Rated specifications

Welding power sources

Model		YD-350GR3Y**	YD-500GR3Y**
			
Number of phase	-	3	
Rated frequency	Hz	50/60	
Rated input	kVA	17	28
	kW	16	26.5
Rated output current	A	DC350	DC500
Rated output voltage	V	DC36	DC45
Rated duty cycle	%	60	100
Adjustable range of output current	A	DC30 - 350	DC60 - 500
Adjustable range of output voltage	V	DC12 - 36	DC14 - 45
Control method	-	IGBT inverter	
Memory function	-	9 ch	
Adjustable welding wire diameter	mm	0.8 / 0.9 / 1.0 / 1.2	1.2 / 1.4 / 1.6
Preflow time	S	0.0 - 5.0 (adjustable by 0.1 sec.)	
Postflow time	S	0.0 - 5.0 (adjustable by 0.1 sec.)	
Arc spot time	S	0.3 - 10.0 (adjustable by 0.1 sec.)	
Dimensions (W) x (D) x (H)	mm	380 x 540* x 640	380 x 540* x 810
Weight	kg	45	65

*Input terminal cover is not included.



We provide the products friendly to the environment.

Aiming for the earth-friendly company, Panasonic Welding Systems Co., Ltd. has been promoting to disuse hazardous substances to the products as the company policy.

Safety precautions ● Before attempting to use any welding product always read the manual to ensure correct use.

Panasonic Welding Systems Co., Ltd.

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